

Research Question

How does temperature stress and desiccation affect the metabolic rate of *Carcinus maenas* and *Hemigrapsus oregonensis*?

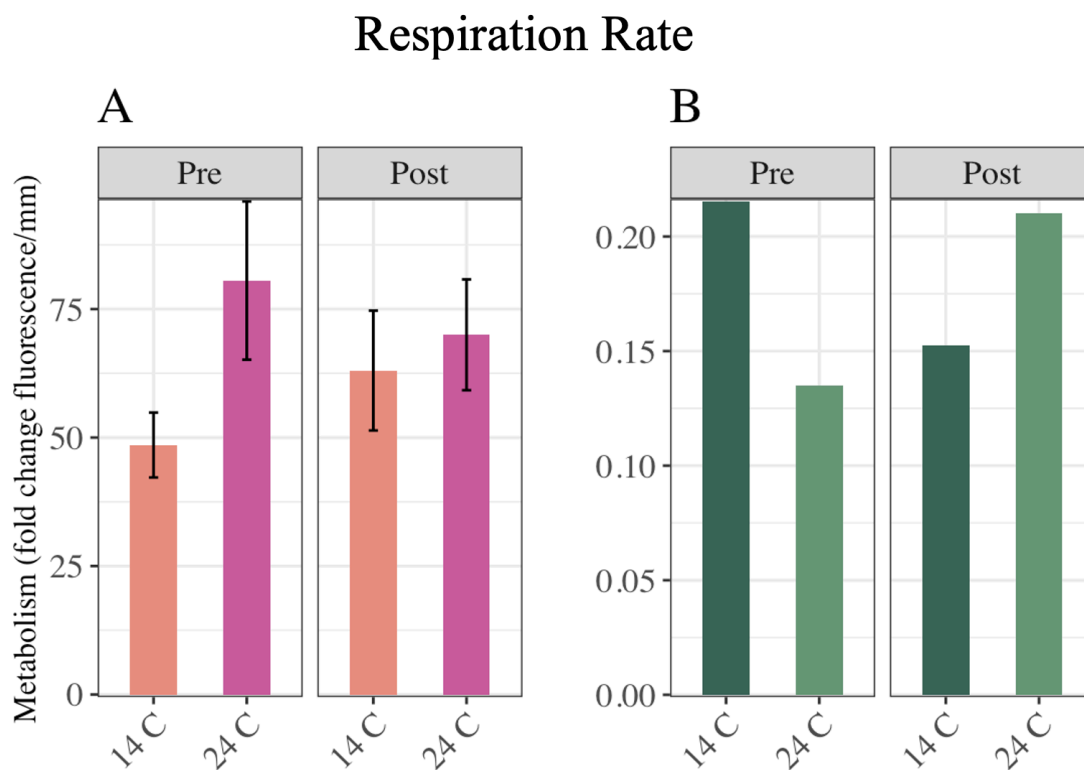
Hypotheses

H₀: Temperature and desiccation will have no impact on crab metabolic rate.

H_{A1}: Metabolic rate will be higher in crabs exposed to elevated water temperatures while in the water and after desiccation.

H_{A2}: *Carcinus maenas* will have less extreme response to the stressors than *Hemigrapsus oregonensis*.

Key Figures & Results



Water Temperature

Figure 1: Average metabolism of *Hemigrapsus oregonensis* (A) and *Carcinus maenas* (B) as fold change in fluorescence/mm at different water temperatures, Pre- and Post- desiccation.

Takeaway: In most cases, metabolism was higher at higher temperatures. In both cases, metabolism was lower after desiccation.

Change in Respiration After Desiccation

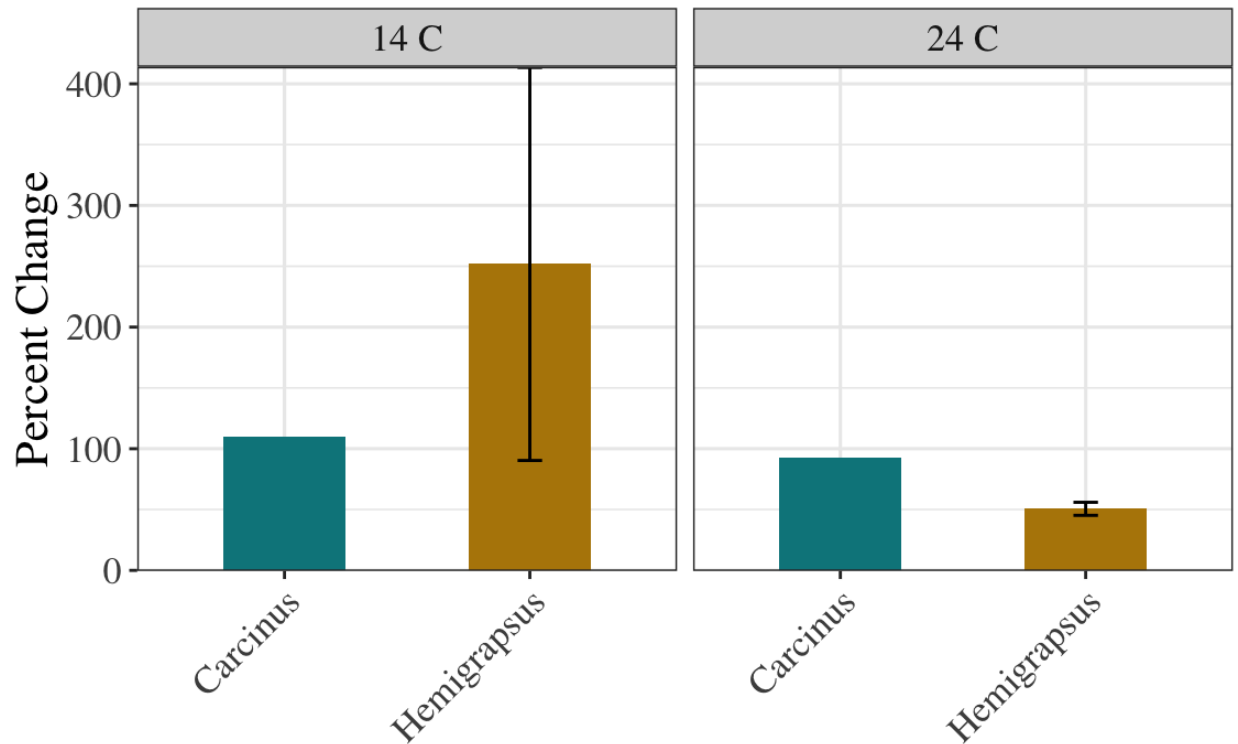


Figure 2: Percent change in respiration rate after desiccation for crabs held at 14 C (left) and 24 C (right).

Takeaway:

Carcinus maenus experienced little change in their metabolism following desiccation.

Table 1. T test of mean metabolism pre and post desiccation for four different experimental groups based on species and temperature.

Species	Temperature	t	df	CI	p.value
Hemigrapsus oregonensis	14 C	0.91	10.44	(-9.69, 23.31)	0.3812629
Hemigrapsus oregonensis	24 C	1.52	11.17	(-2.9, 16.0)	0.1556661
Carcinus maenas	14 C	0.91	1.01	(-22.5, 26.1)	0.5273347
Carcinus maenas	24 C	-1.00	1.00	(-255, 218)	0.5000000

Takeaway: There is no statistically significant difference in metabolism before and after desiccation for any of our experimental groups. Notably, since degrees of freedom (df) is so low for every group this would raise the p-value, and we may see lower p-values after including our replicate data from this week's experiment.

Table 2. Analysis of Variance (ANOVA) test for species and temperature effects on metabolism. Species include *Hemigrapsus oregonensis* and *Carcinus maenas*, temperature is 14 C and 24 C.

Parameter	df	Sum.Sq	Mean.Sq	F.value	p.value
Species	1	41	40.59	0.220	0.641
Temp	1	8	8.10	0.044	0.835
Residuals	45	8291	184.25		

Takeaway: There was no significant difference in metabolism based on species or temperature. With a p value of 0.5 only for the 24C *carcinus maenus* group, it is difficult to obtain statistical significance from this data.